

**AMENDMENT NO. 1 MARCH 2010
TO
IS 14296 : 1995 DOLOMITE FOR REFRACTORY
INDUSTRY — SPECIFICATION**

(Page 1, clause 4) — Delete and renumber the other clauses.

(Page 1, Table 1) — Substitute the following for the existing table:

Table 1 Chemical Composition of Raw Dolomite

Constituent	Requirements, Percent
CaO, <i>Min</i>	28
MgO, <i>Min</i>	21
SiO ₂ , <i>Max</i>	0.6
Fe ₂ O ₃ + Al ₂ O ₃ , <i>Max</i>	0.8

(Page 1, clause 6.2) — Substitute the following for the existing:

‘6.2 Decrepitation Test — Decrepitation test of dolomite and limestone shall be carried out as per Annex A.’

(Page 1, clause 7) — Insert the following Annex A at the end:

ANNEX A
(Clause 6.2)

DECREPITATION TEST FOR DOLOMITE AND LIMESTONE

A-1 Heat causes expansion in crystal matrix of limestones and dolomites. In highly crystalline limestones the heat produces stresses in the individual crystals, causing them to fracture. This is termed as decrepitation. Degree of decrepitation can be determined by the following test.

A-2 SAMPLING

A-2.1 Quantity, two kg of size range from 25 mm to 30 mm.

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A-3 APPARATUS

A-3.1 Muffle Furnace of Size, capable of attaining temperature up to 1 050°C.

A-3.2 Sieve-shaker with sieves of Tyler 4 mesh (4.75 mm opening) and 10 mm opening mesh.

A-3 PROCEDURE

Weigh 2 kg raw dolomite or limestone of required size in a refractory container. Temperature of the furnace is raised to 1 000°C. In the hot furnace refractory container with dolomite/limestone is quickly introduced so that furnace temperature does not drop below 950°C. When the furnace achieves 1 000°C after the drop in temperature (this should not be more than 15 min) the material is allowed to soak for two and half hours in an oxidizing atmosphere. After soaking the material is taken out of the furnace and allowed to cool to ambient temperature. The cold material is then abraded on a 4 mesh in sieve-shaker for 10 minutes. After abrasion fired dolomite/limestone is sieved through 10 mm mesh. The material below 10 mm size is weighed for determining the decrepitation tendency.

A-4 REQUIREMENTS

Quantity of material below 10 mm size shall be less than 10 percent by weight in calcined basis.

(MTD 13)

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